

Genetic report for longevity

Carrier Status

Recessive diseases: alpha-1 antitrypsin (PiMZ), hemochromatosis (HFE H63D), cystic fibrosis, hemoglobinopathies, GJB2 deafness, Mediterranean fever, metabolic errors and what the chip cannot read

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Being a recessive disease is to have a copy of a pathogenic variant and a normal copy. Almost always does not give disease to the own; it matters mainly for the children (if the other parent is carrying the same gene, the risk for pregnancy is 25%). There are exceptions in which a copy already counts a little, and two of them appear here: the alpha-1 antitrypsin PiMZ and, to a lesser extent, the HFE H63D. This report reads the most frequent carrier variants that the chip and the imputation can read, joins the screening of ClinVar already done in the profile and says what you cannot read. The chip does not replace a clinical carrier panel: a clinical panel sequence the entire genes (centenarians of variants per gene, including deletions and repetitions), while here we read one or few variants per gene.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

45 variants	42 in the chip	0 imputed (DR2≥0.8)	3 not found	0 Scores with percentile
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Overall reading. Of the 45 variants of carrier searched, 42 were read. There are two known carrier states and none new. Alpha-1 antitrypsin PiMZ (SERPINA1 Z C/T, without S allele): is the finding with the most consequence, because MZ has some effect on itself (liver with fat or alcohol; lung with tobacco or tobacco) and because each child has a 50% probability of inheriting Z. If the partner also carries Z, the risk of ZZ per pregnancy is 25%. HFE H63D in heterozygosity (C/G), without C282Y or S65C: alone almost never gives iron overload; it would only matter to children if the other person had C282Y. Non-carrier in the variants treated with cystic fibrosis (N1303K, W1282X, G551D, R117H, R553X, 621+1G>T), beta-thalassemia and hemoglobin S and C, G6PD (normal), phenylketonuria, MCAD, hereditary intolerance to fructose, biotinidase, Gaucher N370S, Wilson H1069Q, Smith-Lemli-Opitz, CPT2, McArdle, AMPD1, pseudocholinesterase (normal BCHE), deafness GJB2 (including 35delG by imputation), Mediterranean fever and MUTYH. UGT1A1 *1/*28: a copy of Gilbert's allele (without judgment). FLG: a rare nonsense variant reads as heterozygous, but reading is ambiguous (without judgment). Do not deal with trust: CFTR F508del (without probe; imputation gives 0 copies, but below the confidence threshold), CFTR G542X and FLG R501X. The chip does not replace a clinical carrier panel: a negative result here does not exclude being a carrier.

Favourable

- **SERPINA1 (Pi*S) rs17580 TT:** SERPINA1 Glu264Val (Pi*S): light allele; SZ gives moderate deficit
- **CFTR (N1303K) rs80034486 CC:** CFTR N1303K: serious missense.
- **CFTR (W1282X) rs77010898 GG:** CFTR W1282X: nonsense.
- **CFTR (G551D) rs75527207 GG:** CFTR G551D: channel opening variant (the first treated with ivacaftor).
- **CFTR (R117H) rs78655421 GG:** CFTR R117H: slight variant; the gravity depends on the poly-T (5T/7T) of the 9 intron, which is not read by chip
- **CFTR (R553X) rs74597325 CC:** CFTR R553X: nonsense.
- **CFTR (621+1G>T) rs78756941 GG:** CFTR 621+1G>T: splicing variant.
- **HFE (C282Y) rs1800562 GG:** HFE C282Y: the main variant of hemochromatosis (C282Y/C282Y)
- **HBB (HbS) rs334 TT:** HBB Glu6Val (HbS): sickle cell trait in heterozygosity; sickle cell disease in homozygous.
- **HBB (HbC) rs33930165 CC:** HBB Glu6Lys (HbC).
- **HBB (IVS-I-110) rs35004220 CC:** HBB IVS-I-110 G>A: of the most frequent beta-thalassemia variants.
- **HBB (codon 39) rs11549407 GG:** HBB codon 39 (Gln39Ter, beta-zero).
- **HBB (IVS-I-1) rs33971440 CC:** HBB IVS-I-1 G>A (beta-zero).
- **HBB (IVS-I-6) rs35724775 AA:** HBB IVS-I-6 T>C (beta-plus, slight).
- **G6PD (A-) rs1050828 C:** G6PD Val68Met (A- variant component)
- **G6PD (c.563C>T) rs5030868 G:** G6PD Ser188Phe (severe disability; favism).
- **PAH (R408W) rs5030858 GG:** PAH R408W: the most frequent variants of phenylketonuria.
- **PAH (IVS10-11G>A) rs5030855 CC:** PAH IVS10-11G>A: frequent phenylketonuria variant.
- **ACADM (K329E) rs77931234 AA:** ACADM K329E: ~80% of the MCAD deficit alleles (oxidation of medium chain fatty acids).
- **ALDOB (A149P) rs1800546 CC:** ALDOB A149P: the most frequent variant of hereditary intolerance to fructose.
- **BTBD (D444H) rs13078881 GG:** BTBD D444H: partial biotinidase deficit allele (frequent, ~4%); only results in combination with a severe allele.
- **GBA1 (N370S) rs76763715 TT:** GBA1 N370S: the most frequent variant of Gaucher's disease; in heterozygosity also increases the risk of Parkinson's disease
- **ATP7B (H1069Q) rs76151636 GG:** ATP7B H1069Q: the most frequent variant of Wilson's disease (copper accumulation).

- **DHCR7 (W151X) rs11555217 CC:** DHCR7 W151X: Smith-Lemli-Opitz syndrome (cholesterol synthesis).
- **CPT2 (S113L) rs74315294 CC:** CPT2 S113L: the most frequent variant of muscle CPT2 deficit (pain and dark urine with prolonged exercise, fasting or cold).
- **AMPD1 (Q12X) rs17602729 GG:** AMPD1 Q12X: very frequent (~10% of carriers); in homozygosity, myoadenylate deaminase deficiency, often without symptoms
- **PYGM (R50X) rs116987552 GG:** PYGM R50X: the most frequent variant of McArdle's disease (intolerant to intense exercise).
- **BCHE (variant A) rs1799807 TT:** BCHE D70G (atypical variant): in homozygosity, prolonged apnea with succinylcholine and mivacurium (anesthesia).
- **BCHE (Kvariant) rs1803274 CC:** BCHE A539T (Kvariant): reduces ~30% activity; only the variant A.
- **GJB2 (35delG) rs80338939 without deletion:** GJB2 35delG: one of the most frequent variants of congenital deafness
- **GJB2 (M34T) rs35887622 AA:** GJB2 M34T: light allele, low penetration.
- **GJB2 (V37I) rs72474224 CC:** GJB2 V37I: light allele (light deafness in homozygosity).
- **GJB2 (L90P) rs80338945 AA:** GJB2 L90P.
- **MEFV (M694V) rs61752717 TT:** MEFV M694V: the most severe variant of familial Mediterranean fever.
- **MUTYH (Y179C) rs34612342 TT:** MUTYH Y179C: one of the two most frequent variants of polyposis associated with MUTYH
- **MUTYH (G396D) rs36053993 CC:** MUTYH G396D: the other frequent variant of polyposis associated with MUTYH.

Unfavourable

- **SERPINA1 (Pi*Z) rs28929474 CT:** SERPINA1 Glu342Lys (Pi*Z): a copy = PiMZ, with antitrypsin alpha-1 deficiency
- **HFE (H63D) rs1799945 CG:** HFE H63D: a copy, already known in the profile

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 42 chip genotypes, 0 needed complement, 3 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Lung and liver: alpha-1 antitrypsin and cystic fibrosis

SERPINA1: Z and S alleles produce alpha-1 antitrypsin that folds poorly; Z polymerizes within the liver cells (protein aggregation) and lacks in the lung. ZZ is the disease; MZ (one copy) has about 60% of the protein and moderately increased risk of liver disease with other factors (liver fat, alcohol) and emphysema with tobacco or tobacco. CFTR: the most frequent variants of cystic fibrosis that the chip reads; F508del is a deletion and only reads if the chip has its own probe.

Ageing mechanisms: Proteostasis loss; Protein aggregation; Chronic Inflammation

Favourable

- SERPINA1 (Pi*S) rs17580 TT
- CFTR (N1303K) rs80034486 CC
- CFTR (W1282X) rs77010898 GG
- CFTR (G551D) rs75527207 GG
- CFTR (R117H) rs78655421 GG
- CFTR (R553X) rs74597325 CC
- CFTR (621+1G>T) rs78756941 GG

Unfavourable

- SERPINA1 (Pi*Z) rs28929474 CT

Favourable variants (7)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SERPINA1 (Pi*S)	rs17580 14:94380925 · T/A	TT	chip	A (0; 6%)	strong	SERPINA1 Glu264Val (Pi*S): light allele; SZ gives moderate deficit. Without the S allele, the genotype is MZ (not SZ).
CFTR (N1303K)	rs80034486 7:117652877 · C/G	CC	chip	G (0; 0%)	strong	CFTR N1303K: serious missense.
CFTR (W1282X)	rs77010898 7:117642566 · G/A/C	GG	chip	A (0; —)	strong	CFTR W1282X: nonsense
CFTR (G551D)	rs75527207 7:117587806 · G/A/T	GG	chip	A (0; —)	strong	CFTR G551D: channel opening variant (the first treated with ivacaftor).
CFTR (R117H)	rs78655421 7:117530975 · G/A/C/T	GG	chip	A (0; —)	moderate	CFTR R117H: slight variant; the gravity depends on the poly-T (5T/7T) of the 9 intron, which is not read by chip. It is associated with the congenital absence of different channels.
CFTR (R553X)	rs74597325 7:117587811 · C/G/T	CC	chip	T (0; 0%)	strong	CFTR R553X: nonsense
CFTR (621+1G>T)	rs78756941 7:117531115 · G/A/T	GG	chip	T (0; —)	strong	CFTR 621+1G>T: splicing variant.

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SERPINA1 (Pi*Z)	rs28929474 14:94378610 · C/G/T	CT	chip	T (1; 2%)	strong	SERPINA1 Glu342Lys (Pi*Z): a copy = PiMZ, with antitrypsin alpha-1 deficiency. Already known in the profile and in the Liver and Lung reports. Unlike most carrier states, MZ has some effect on its own: moderately increased risk of liver fibrosis with fat in the liver or alcohol and emphysema with tobacco or tobacco.

SNP not found in the data (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CFTR (F508del)	rs113993960 7:117559591 · TCTT/T/ TCTTCTT	—	SNP not found in the data (without probe on the chip; imputed with DR2=0.68 < 0.8)	—	strong	CFTR F508del: the most frequent variant of cystic fibrosis (~70% of alleles). It is a deletion of 3 bases; only if the chip has its own probe. Imputation of deletion gives 0 copies (0 dosage), but with DR2=0.68 does not reach the confidence threshold. Probably non-carrier; only one clinical test confirms it.
CFTR (G542X)	rs113993959 7:117587778 · G/A/T	—	SNP not found in the data	T (—; —)	strong	CFTR G542X: nonsense; the most frequent.

Blood and iron: HFE, hemoglobins and G6PD

HFE: C282Y/C282Y is the genotype of hemochromatosis; C282Y/H63D gives light iron overload in some; H63D alone (one copy) is very common (~25% of people) and has a small effect on iron markers. HBB: sickle cell disease (HbS), HbC and the most common beta-thalassemia variants. G6PD is in the X chromosome: in a man there is no "carrier", or has or does not have the disability.

Ageing mechanisms: Oxidative stress; Redox imbalance

Favourable

- HFE (C282Y) rs1800562 GG
- HFE (S65C) rs1800730 AA
- HBB (HbS) rs334 TT
- HBB (HbC) rs33930165 CC
- HBB (IVS-I-110) rs35004220 CC
- HBB (codon 39) rs11549407 GG
- HBB (IVS-I-1) rs33971440 CC
- HBB (IVS-I-6) rs35724775 AA
- G6PD (A-) rs1050828 C
- G6PD (c.563C>T) rs5030868 G

Unfavourable

- HFE (H63D) rs1799945 CG

Favourable variants (10)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
HFE (C282Y)	rs1800562 6:26092913 · G/A	GG	chip	A (0; 4%)	strong	HFE C282Y: the main variant of hemochromatosis (C282Y/C282Y). No copies.
HFE (S65C)	rs1800730 6:26090957 · A/T	AA	chip	T (0; 2%)	weak	HFE S65C: effect even lighter than H63D.
HBB (HbS)	rs334 11:5227002 · T/A/C/G	TT	chip	A (0; 0%)	strong	HBB Glu6Val (HbS): sickle cell trait in heterozygosity; sickle cell disease in homozygous.
HBB (HbC)	rs33930165 11:5227003 · C/A/G/T	CC	chip	T (0; 0%)	strong	HBB Glu6Lys (HbC).
HBB (IVS-I-110)	rs35004220 11:5226820 · C/T	CC	chip	T (0; —)	strong	HBB IVS-I-110 G>A: of the most frequent beta-thalassemia variants.
HBB (codon 39)	rs11549407 11:5226774 · G/A/C/T	GG	chip	A (0; 0%)	strong	HBB codon 39 (Gln39Ter, beta-zero).
HBB (IVS-I-1)	rs33971440 11:5226929 · C/A/G/T	CC	chip	T (0; 0%)	strong	HBB IVS-I-1 G>A (beta-zero).
HBB (IVS-I-6)	rs35724775 11:5226924 · A/G/T	AA	chip	G (0; —)	strong	HBB IVS-I-6 T>C (beta-plus, slight).
G6PD (A-)	rs1050828 X:154536002 · C/T	C	chip	T (0; 0%)	strong	G6PD Val68Met (A- variant component). Chromosome X: in a man, a copy = disability.
G6PD (c.563C>T)	rs5030868 X:154534419 · G/A	G	chip	A (0; 0%)	strong	G6PD Ser188Phe (severe disability; favism).

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Evidence	Interpretation
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				Unfavourable allele (copying; frequency in reference)		
HFE (H63D)	rs1799945 6:26090951 · C/G/T	CG	chip	G (1; 17%)	moderate	HFE H63D: a copy, already known in the profile. On its own, it almost never gives iron overload (sobe a little the transferrin saturation and ferritin on average); only counts in combination with C282Y (composite genome C282Y/H63D), which does not exist here. For children: if the other person has C282Y, ~25% of C282Y/H63D.

Metabolism errors and enzymes

Phenylketonuria (PAH), MCAD deficiency (ACADM, oxidation of medium chain fatty acids), hereditary intolerance to fructose (ALDOB), biotinidase deficit (BTD), Gaucher disease (GBA1), Wilson's disease (ATP7B), Smith-Lemli-Opitz (DHCR7), CPT2 and myoadenylate deaminase (AMPD1), McArdle's disease (PYGM), pseudocholinesterase (BCHE, anesthesia) and Gilbert (UGT1A1).

Ageing mechanisms: Metabolic Homeostasis decline; Mitochondrial dysfunction; Lysosome dysfunction

Favourable

Unfavourable

- PAH (R408W) rs5030858 GG
 - PAH (IVS10-11G>A) rs5030855 CC
 - ACADM (K329E) rs77931234 AA
 - ALDOB (A149P) rs1800546 CC
 - BTD (D444H) rs13078881 GG
 - GBA1 (N370S) rs76763715 TT
 - ATP7B (H1069Q) rs76151636 GG
 - DHCR7 (W151X) rs11555217 CC
 - CPT2 (S113L) rs74315294 CC
 - AMPD1 (Q12X) rs17602729 GG
 - PYGM (R50X) rs116987552 GG
 - BCHE (variant A) rs1799807 TT
 - BCHE (Kvariant) rs1803274 CC
- None.

Favourable variants (13)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
PAH (R408W)	rs5030858 12:102840493 · G/A	GG	chip	A (0; 0%)	strong	PAH R408W: the most frequent variants of phenylketonuria.
PAH (IVS10-11G>A)	rs5030855 12:102843790 · C/T	CC	chip	T (0; 0%)	strong	PAH IVS10-11G>A: frequent phenylketonuria variant.
ACADM (K329E)	rs77931234 1:75761161 · A/C/G	AA	chip	G (0; 0%)	strong	ACADM K329E: ~80% of the MCAD deficit alleles (oxidation of medium chain fatty acids).
ALDOB (A149P)	rs1800546 9:101427574 · C/G/T	CC	chip	G (0; 1%)	strong	ALDOB A149P: the most frequent variant of hereditary intolerance to fructose.
BTD (D444H)	rs13078881 3:15645186 · G/C/T	GG	chip	C (0; 4%)	moderate	BTD D444H: partial biotinidase deficit allele (frequent, ~4%); only results in combination with a severe allele.
GBA1 (N370S)	rs76763715 1:155235843 · T/C/G	TT	chip	C (0; 0%)	strong	GBA1 N370S: the most frequent variant of Gaucher's disease; in heterozygosity it also increases the risk of Parkinson's disease. GBA1 has a very similar pseudogene, so chip readings fail easily.
ATP7B (H1069Q)	rs76151636 13:51944145 · G/A/T	GG	chip	T (0; 0%)	strong	ATP7B H1069Q: the most frequent variant of Wilson's disease (copper accumulation).
DHCR7 (W151X)	rs11555217 11:71441401 · C/G/T	CC	chip	T (0; —)	strong	DHCR7 W151X: Smith-Lemli-Opitz syndrome (cholesterol synthesis).
CPT2 (S113L)	rs74315294 1:53202427 · C/T	CC	chip	T (0; 0%)	strong	CPT2 S113L: the most frequent variant of muscle CPT2 deficit (pain and dark urine with prolonged exercise, fasting or cold).

AMPD1 (Q12X)	rs17602729 1:114693436 · G/A/T	GG	chip	A (0; 12%)	moderate	AMPD1 Q12X: very frequent (~10% of carriers); in homozygosity, myoadenylate deaminase deficiency, often without symptoms. I also read in the Sports Profile report.
PYGM (R50X)	rs116987552 11:64759751 · G/A/C	GG	chip	A (0; 0%)	strong	PYGM R50X: the most frequent variant of McArdle's disease (intolerant to intense exercise).
BCHE (variant A)	rs1799807 3:165830741 · T/C	TT	chip	C (0; 2%)	strong	BCHE D70G (atypical variant): in homozygosity, prolonged apnea with succinylcholine and mivacurium (anesthesia).
BCHE (variant K)	rs1803274 3:165773492 · C/T	CC	chip	T (0; 19%)	moderate	BCHE A539T (Kvariant): reduces ~30% activity; only the variant A.

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
UGT1A1 (*28)	rs887829 2:233759924 · C/T	CT	chip	—	strong	UGT1A1*28 (rs887829 marker): C/T = *1/*28, Gilbert allele carrier (already in the profile). Gilbert is *28/*28: indirect bilirubin a little high, benign. A copy does not give Gilbert.

Hearing, skin, inflammation and recessive cancer

GJB2 (connectin 26, the most frequent genetic cause of congenital deafness), FLG (filaggrin, skin barrier), MEFV (familial Mediterranean fever) and MUTYH (polyposis associated with MUTYH, recessive: only two copies give high risk of colon cancer).

Ageing mechanisms: Chronic Inflammation; DNA Damage and DNA Repair Deficit

Favourable	Unfavourable
• GJB2 (35delG) rs80338939 without deletion • GJB2 (M34T) rs35887622 AA • GJB2 (V37I) rs72474224 CC • GJB2 (L90P) rs80338945 AA • MEFV (M694V) rs61752717 TT • MEFV (E148Q) rs3743930 CC • MUTYH (Y179C) rs34612342 TT • MUTYH (G396D) rs36053993 CC	None.

Favourable variants (8)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
GJB2 (35delG)	rs80338939 13:20189547 · CCCCCC/CCCCC/CCCCCC	without deletion	imputed DR2=0.96 (deletion) + chip probe	35delG (0; —)	strong	GJB2 35delG: one of the most frequent variants of congenital deafness. It is a deletion; it only reads with its own probe. The imputation of the deletion gives 0 copies with DR2=0.96 (0 dosage) and the probe owns the chip is homozygous (AA), without the heterozygosity of a carrier.
GJB2 (M34T)	rs35887622 13:20189481 · A/C/G	AA	chip	G (0; 2%)	moderate	GJB2 M34T: light allele, low penetration.
GJB2 (V37I)	rs72474224 13:20189473 · C/A/T	CC	chip	T (0; 0%)	moderate	GJB2 V37I: light allele (light deafness in homozygosity).
GJB2 (L90P)	rs80338945 13:20189313 · A/G	AA	chip	G (0; —)	strong	GJB2 L90P.
MEFV (M694V)	rs61752717 16:3243407 · T/A/C	TT	chip	C (0; 0%)	strong	MEFV M694V: the most severe variant of familial Mediterranean fever.
MEFV (E148Q)	rs3743930 16:3254626 · C/A/G/T	CC	chip	G (0; 1%)	weak	MEFV E148Q: frequent and very low penetrance; many laboratories already consider it a variant of uncertain significance.
MUTYH (Y179C)	rs34612342 1:45332803 · T/C	TT	chip	C (0; 0%)	strong	MUTYH Y179C: one of the two most frequent variants of polyposis associated with MUTYH. A copy: risk of colon cancer when very slightly increased.
MUTYH (G396D)	rs36053993 1:45331556 · C/A/T	CC	chip	T (0; 1%)	strong	MUTYH G396D: the other frequent variant of polyposis associated with MUTYH.

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
		GC	chip	C (1; 0%)	weak	

FLG rs542799026
(nonsense 1:152306769 ·
rare) G/A/C

FLG: rare nonsense variant reads as heterozygous in the profile ClinVar screening ("possible, unconfirmed"). The FLG has internal repetitions where the chips fail frequently. Classified without judgment: ambiguous reading by chip (very rare variant, gene with repetitions). Only FLG sequencing would confirm.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FLG (R501X)	rs61816761 1:152313385 · G/A/C/T	—	SNP not found in the data (Imputed with DR2=0.48 < 0.8)	A (—; 1%)	strong	FLG R501X: the most frequent filaggrin loss variant (dry skin, ichthyosis vulgar, atopic dermatitis; semi-dominant effect). Not dealing with trust (FLG is difficult to read by chip and imputation).

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Protein Aggregation	SERPINA1 (Pi*Z) CT	SERPINA1 (Pi*S) TT; CFTR (N1303K) CC; CFTR (W1282X) GG; CFTR (G551D) GG; CFTR (R117H) GG; CFTR (R553X) CC; CFTR (621+1G>T) GG	—
Redox imbalance	HFE (H63D) CG	HFE (C282Y) GG; HFE (S65C) AA; HBB (HbS) TT; HBB (HbC) CC; HBB (IVS-I-110) CC; HBB (codon 39) GG; HBB (IVS-I-1) CC; HBB (IVS-I-6) AA; G6PD (A-) C; G6PD (c.563C>T)	—
Chronic Inflammation	SERPINA1 (Pi*Z) CT	SERPINA1 (Pi*S) TT; CFTR (N1303K) CC; CFTR (W1282X) GG; CFTR (G551D) GG; CFTR (R117H) GG; CFTR (R553X) CC; CFTR (621+1G>T) GG; GJB2 (35delG) without deletion; GJB2 (M34T) AA; GJB2 (V37I) CC; GJB2 (L90P) AA; MEFV (M694V) TT; MEFV (E148Q) CC; MUTYH (Y179C) TT; MUTYH (G396D) CC	—
Proteostasis loss	SERPINA1 (Pi*Z) CT	SERPINA1 (Pi*S) TT; CFTR (N1303K) CC; CFTR (W1282X) GG; CFTR (G551D) GG; CFTR (R117H) GG; CFTR (R553X) CC; CFTR (621+1G>T) GG	—
Oxidative stress	HFE (H63D) CG	HFE (C282Y) GG; HFE (S65C) AA; HBB (HbS) TT; HBB (HbC) CC; HBB (IVS-I-110) CC; HBB (codon 39) GG; HBB (IVS-I-1) CC; HBB (IVS-I-6) AA; G6PD (A-) C; G6PD (c.563C>T)	—
DNA Damage and DNA Repair Deficit	—	GJB2 (35delG) without deletion; GJB2 (M34T) AA; GJB2 (V37I) CC; GJB2 (L90P) AA; MEFV (M694V) TT; MEFV (E148Q) CC; MUTYH (Y179C) TT; MUTYH (G396D) CC	—
Metabolic Homeostasis Decline	—	PAH (R408W) GG; PAH (IVS10-11G>A) CC; ACADM (K329E) AA; ALDOB (A149P) CC; BTBD (D444H) GG; GBA1 (N370S) TT; ATP7B (H1069Q) GG; DHCR7 (W151X) CC; CPT2 (S113L) CC; AMPD1 (Q12X) GG; PYGM (R50X) GG; BCHE (variant A) TT; BCHE (K variant) CC	—
Lysosome dysfunction	—	PAH (R408W) GG; PAH (IVS10-11G>A) CC; ACADM (K329E) AA; ALDOB (A149P) CC; BTBD (D444H) GG; GBA1 (N370S) TT; ATP7B (H1069Q) GG; DHCR7 (W151X) CC; CPT2 (S113L) CC; AMPD1 (Q12X) GG; PYGM (R50X) GG; BCHE (variant A) TT; BCHE (K variant) CC	—
Mitochondrial dysfunction	—	PAH (R408W) GG; PAH (IVS10-11G>A) CC; ACADM (K329E) AA; ALDOB (A149P) CC; BTBD (D444H) GG; GBA1 (N370S) TT; ATP7B (H1069Q) GG; DHCR7 (W151X) CC; CPT2 (S113L) CC; AMPD1 (Q12X) GG; PYGM (R50X) GG; BCHE (variant A) TT; BCHE (K variant) CC	—

Practical part

Ideas related to carrier states, crossed with the genetic plan v2 and current medication (isotretinoin, rapamycin, retatrutide, dapagliflozin, telmisartan, tadalafil, lithium 5 mg, naltrexone in low dose). Everything is suggestion (not applied).

1. Clinical patient panel before planning children

Genetics: SERPINA1 PiMZ; HFE H63D; F508del and other unread variants

If there are plans of children, make a panel of clinical carrier (sequencing) for the two members of the couple, informing the laboratory of PiMZ and H63D already known. The chip does not replace this panel.

V2 genetic plan and medication: No change in the plan. • suggestion (not applied)

2. Confirm the PiMZ and inform the family

Genetics: SERPINA1 Z C/T

Confirm with the alpha-1 antitrypsin trial and the clinical phenotype or Pi genotype, if not yet done. Parents, siblings and children have 50% probability of having Z; they can make the same simple test.

V2 genetic plan and medication: No change in the plan. • suggestion (not applied)

3. Protecting the liver and lung

Genetics: SERPINA1 PiMZ

The risk of MZ is concentrated in the liver with fat or alcohol and in the lung with tobacco or tobacco. Keeping alcohol at zero and the liver without fat, avoiding tobacco, vaping and occupational smoking. During isotretinoin and rapamycin, routine liver analyses gain more value.

V2 genetic plan and medication: Compatible with the plan (zero alcohol, already in progress). • suggestion (not applied)

4. Iron: ferritin objective of the plan

Genetics: HFE H63D heterozygous, without C282Y

H63D alone has a small effect, but it is added to other causes of high iron. Keep the plan objective (ferritin below 100) and do not take iron supplements without indication.

V2 genetic plan and medication: Compatible with the plan (ferritin < 100). • suggestion (not applied)

Avoiding or with caution

- Tobacco, vaping, passive smoke and occupational dust (PiMZ). suggestion (not applied)
- Iron supplements without confirmed deficit (H63D). suggestion (not applied)
- Taking a "non-carrier" of the chip as a guarantee: only one clinical panel confirms it. suggestion (not applied)

Limitations

- **The chip does not replace a clinical carrier panel.** A clinical panel (sequencing of tens to hundreds of genes, with deletions and duplications) reads thousands of variants per gene; here we read one to eight variants in the most frequent genes.
- They do not read by chip: spinal muscle atrophy (SMN1, number of copies), congenital adrenal hyperplasia (CYP21A2, pseudogene and conversions), alpha-thalassemia (HBA1/HBA2) and fragile X (FMR1 repeats, relevant only for women). F8 inversions of hemophilia A and most small deletions and insertions (e.g. CFTR F508del and HEXA 1278insTATC) when there is no probe itself.
- Rare variants: chips have a high false positive rate for very rare variants, and rare variants are not reliable. A positive result has to be confirmed by a clinical test; a negative result does not exclude being carriers of other variants of the same gene.
- The complete ClinVar tracking of the chip (684 967 markers) found pathogenic or probably pathogenic 14 records; only SERPINA1 Pi*Z is plausible. The others are probable artifacts (homozygosity for severe rare variants in a healthy adult, probes of multiple variants, pseudogenes); see the Hereditary Genetic Diseases report.
- Only genes do not include analysis (e.g., alpha-1 antitrypsin or iron trial). It is not a diagnosis.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local statistical imputation (with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: .

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

- Gregg AR et al. Screening for autosomal recessive and X-linked conditions during pregnancy and preconception: a practice resource of the ACMG (Genet Med 2021; PMID 34285390).
- Strnad P, McElvaney NG, Lomas DA. Alpha1-antitrypsin deficiency (N Engl J Med 2020; PMID 32294348).
- Schneider CV et al. Adult liver phenotypes Pi^*MZ and Pi^*ZZ in the large reference cohort (Gastroenterology 2020).
- Gurrin LC et al. HFE C282Y/H63D composite heterozygotes are at low risk of hemochromatosis-related morbidity (Hepatology 2009).
- Weedon MN et al. Use of SNP chips to detect rare pathogenic variants: retrospective, population based diagnostic evaluation (BMJ 2021; PMID 33589468).
- CFTR2: clinical and functional database of CFTR variants.
- ClinVar (NCBI), 28-09-2026 version.

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